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Kelas : S1 If-13-05

JAWABAN ASESMEN 3 SEARCHING & SORTING

1. Source Code jawaban :

```
2. package main
3.
4. import (
5.     "fmt"
6. )
7.
8. const NMAX = 1000000
9.
10. type arrInt [NMAX]int
11.
12. func SelectionSort(T *arrInt, n int) {
13.     for i := 0; i < n-1; i++ {
14.         minIdx := i
15.         for j := i + 1; j < n; j++ {
16.             if T[j] < T[minIdx] {
17.                 minIdx = j
18.             }
19.         }
20.         T[i], T[minIdx] = T[minIdx], T[i]
21.     }
22. }
23.
24. func median(T arrInt, n int) float64 {
25.     if n == 0 {
26.         return 0
27.     }
28.     if n%2 != 0 {
29.         return float64(T[n/2])
30.     }
31.     return float64(T[n/2-1]+T[n/2]) / 2.0
32. }
33.
34. func main() {
```

```

35.  var A arrInt
36.  var x int
37.  var n int = 0
38.
39.  fmt.Println("Input data masukan :")
40.  fmt.Scan(&x)
41.  for x != -5313541 && n < NMAX {
42.      if x == 0 {
43.          if n > 0 {
44.              SelectionSort(&A, n)
45.              fmt.Println("Median :")
46.              fmt.Println(median(A, n))
47.          }
48.      } else {
49.          A[n] = x
50.          n++
51.      }
52.      fmt.Scan(&x)
53.  }
54. }
55.

```

Screenshot Output :

The screenshot shows a Go IDE with the following components:

- EXPLORER:** Shows the file structure with 'soal1.go' open.
- EDITOR:** Displays the source code for 'soal1.go'. The code includes a package declaration, imports, constants, type definitions, and functions for Selection Sort and median calculation.
- OUTPUT/TERMINAL:** Shows the execution of the program. It prompts for 'Input data masukan :', takes two sets of input, and prints the corresponding 'Median' values.

Source Code (soal1.go):

```

1 package main
2
3 import (
4     "fmt"
5 )
6
7 const NMAX = 1000000
8
9 type arrInt [NMAX]int
10
11 func SelectionSort(T *arrInt, n int) {
12     for i := 0; i < n-1; i++ {
13         minIdx := i
14         for j := i + 1; j < n; j++ {
15             if T[j] < T[minIdx] {
16                 minIdx = j
17             }
18         }
19         T[i], T[minIdx] = T[minIdx], T[i]
20     }
21 }
22
23 func median(T arrInt, n int) float64 {
24     if n == 0 {
25         return 0
26     }
27     if n%2 != 0 {
28         return float64(T[n/2])
29     }
30     return float64(T[n/2-1]+T[n/2]) / 2.0
31 }
32

```

Terminal Output:

```

PS C:\Users\Hp\Documents\1090825_00141_Fauzi-Ulhaq-Abdillah\Week 13 - Asesment 3> go run soal1.go
Input data masukan :
7 23 11 0 5 19 2 29 3 13 17 0 -5313541
Median :
11
Median :
12
PS C:\Users\Hp\Documents\1090825_00141_Fauzi-Ulhaq-Abdillah\Week 13 - Asesment 3> go run soal1.go
Input data masukan :
23 12 24 26 20 10 25 8 0 -5313541
Median :
21.5
Median :
26.5
PS C:\Users\Hp\Documents\1090825_00141_Fauzi-Ulhaq-Abdillah\Week 13 - Asesment 3> go run soal1.go
Input data masukan :
19 21 34 67 32 1 0 45 32 44 76 0 99 89 0 -5313541
Median :
33
Median :
39
PS C:\Users\Hp\Documents\1090825_00141_Fauzi-Ulhaq-Abdillah\Week 13 - Asesment 3>

```

Penjelasan Singkat Cara Kerja Program :

Program diatas adalah sistem program yang secara dinamis mengakumulasi data dan menghitung nilai tengah (median) secara progresif. Program tersebut menggunakan metode selection sort dan menghitung nilai median sesuai dengan kondisi ganjil atau genap

2. Source Code Jawaban :

```
package main

import (
    "fmt"
)

type Pemain struct {
   >NamaDepan    string
   >NamaBelakang string
   >Gol          int
   >Assist       int
}

func main() {
    var n int

    fmt.Println("Masukkan Data Input :")
    fmt.Scan(&n)

    dataPemain := make([]Pemain, n)

    for i := 0; i < n; i++ {
```

```

        fmt.Scan(&dataPemain[i].NamaDepan,
&dataPemain[i].NamaBelakang, &dataPemain[i].Gol,
&dataPemain[i].Assist)

    }

    for i := 1; i < n; i++ {

        key := dataPemain[i]

        j := i - 1

        for j >= 0 && (dataPemain[j].Gol < key.Gol ||
(dataPemain[j].Gol == key.Gol && dataPemain[j].Assist < key.Assist)) {

            dataPemain[j+1] = dataPemain[j]

            j--

        }

        dataPemain[j+1] = key

    }

    fmt.Println("\nHasil Sorting :")

    for i := 0; i < n; i++ {

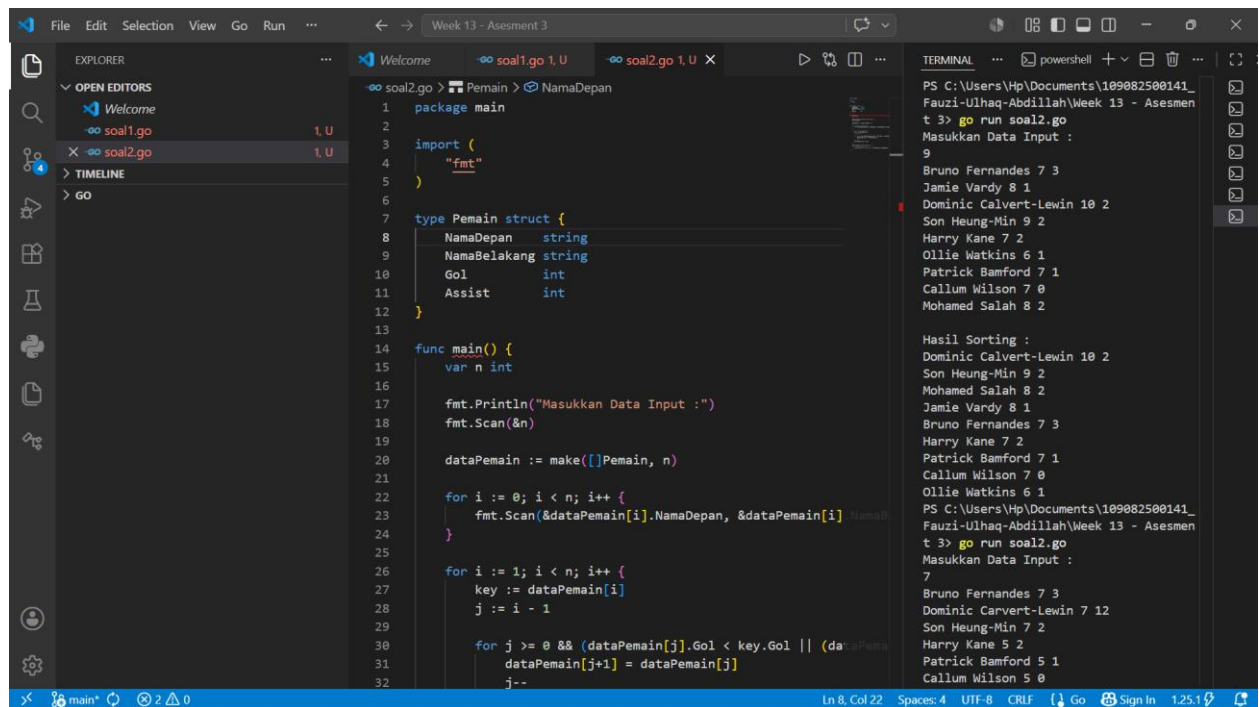
        fmt.Printf("%s %s %d %d\n", dataPemain[i].NamaDepan,
dataPemain[i].NamaBelakang, dataPemain[i].Gol, dataPemain[i].Assist)

    }

}

```

Screenshot Output :



The screenshot shows a Go IDE with two files open: `soal1.go` and `soal2.go`. The `soal2.go` file contains the following code:

```
1 package main
2
3 import (
4     "fmt"
5 )
6
7 type Pemain struct {
8     NamaDepan string
9     NamaBelakang string
10    Gol int
11    Assist int
12 }
13
14 func main() {
15     var n int
16
17     fmt.Println("Masukkan Data Input :")
18     fmt.Scan(&n)
19
20     dataPemain := make([]Pemain, n)
21
22     for i := 0; i < n; i++ {
23         fmt.Scan(&dataPemain[i].NamaDepan, &dataPemain[i].NamaBelakang, &dataPemain[i].Gol, &dataPemain[i].Assist)
24     }
25
26     for i := 1; i < n; i++ {
27         key := dataPemain[i]
28         j := i - 1
29
30         for j >= 0 && (dataPemain[j].Gol < key.Gol || (dataPemain[j].Gol == key.Gol && dataPemain[j].Assist < key.Assist)) {
31             dataPemain[j+1] = dataPemain[j]
32             j--
33         }
34         dataPemain[j+1] = key
35     }
36
37     for i := 0; i < n; i++ {
38         fmt.Println(dataPemain[i].NamaDepan, dataPemain[i].NamaBelakang, dataPemain[i].Gol, dataPemain[i].Assist)
39     }
40 }
```

The terminal output shows the execution of the program. It prompts for the number of players (n) and then for each player's name, last name, goals, and assists. The output shows the sorted list of players based on goals and assists.

```
PS C:\Users\Hp\Documents\109082500141_
Fauzi-Ulhaq-Abdillah\Week 13 - Asesmen
t 3> go run soal2.go
Masukkan Data Input :
9
Bruno Fernandes 7 3
Jamie Vardy 8 1
Dominic Calvert-Lewin 10 2
Son Heung-Min 9 2
Harry Kane 7 2
Ollie Watkins 6 1
Patrick Bamford 7 1
Callum Wilson 7 0
Mohamed Salah 8 2

Hasil Sorting :
Dominic Calvert-Lewin 10 2
Son Heung-Min 9 2
Mohamed Salah 8 2
Jamie Vardy 8 1
Bruno Fernandes 7 3
Harry Kane 7 2
Patrick Bamford 7 1
Callum Wilson 7 0
Ollie Watkins 6 1
PS C:\Users\Hp\Documents\109082500141_
Fauzi-Ulhaq-Abdillah\Week 13 - Asesmen
t 3> go run soal2.go
Masukkan Data Input :
7
Bruno Fernandes 7 3
Dominic Carvert-Lewin 7 12
Son Heung-Min 7 2
Harry Kane 5 2
Patrick Bamford 5 1
Callum Wilson 5 0
```

Penjelasan Singkat Cara Kerja Program :

Program tersebut bekerja menggunakan algoritma Insertion Sort untuk mengurutkan data pemain secara menurun (descending) berdasarkan kriteria utama yaitu jumlah gol, serta kriteria kedua yaitu jumlah assist jika jumlah golnya sama.

3. Source Code Jawaban :

```
package main

import "fmt"

const NMAX = 1000000

type partai struct {
    nama int
    suara int
}
```

```
type tabPartai [NMAX]partai

func posisi(t tabPartai, n int, nama int) int {
    for i := 0; i < n; i++ {
        if t[i].nama == nama {
            return i
        }
    }
    return -1
}

func main() {
    var p tabPartai
    var n int = 0
    var inputNama int

    fmt.Println("Masukkan proses input suara :")
    for {
        fmt.Scan(&inputNama)
        if inputNama == -1 {
            break
        }

        idx := posisi(p, n, inputNama)
        if idx != -1 {
            p[idx].suara++
        } else {
            p[n].nama = inputNama
            p[n].suara = 1
        }
    }
}
```

```

        n++
    }
}

for i := 1; i < n; i++ {
    key := p[i]
    j := i - 1

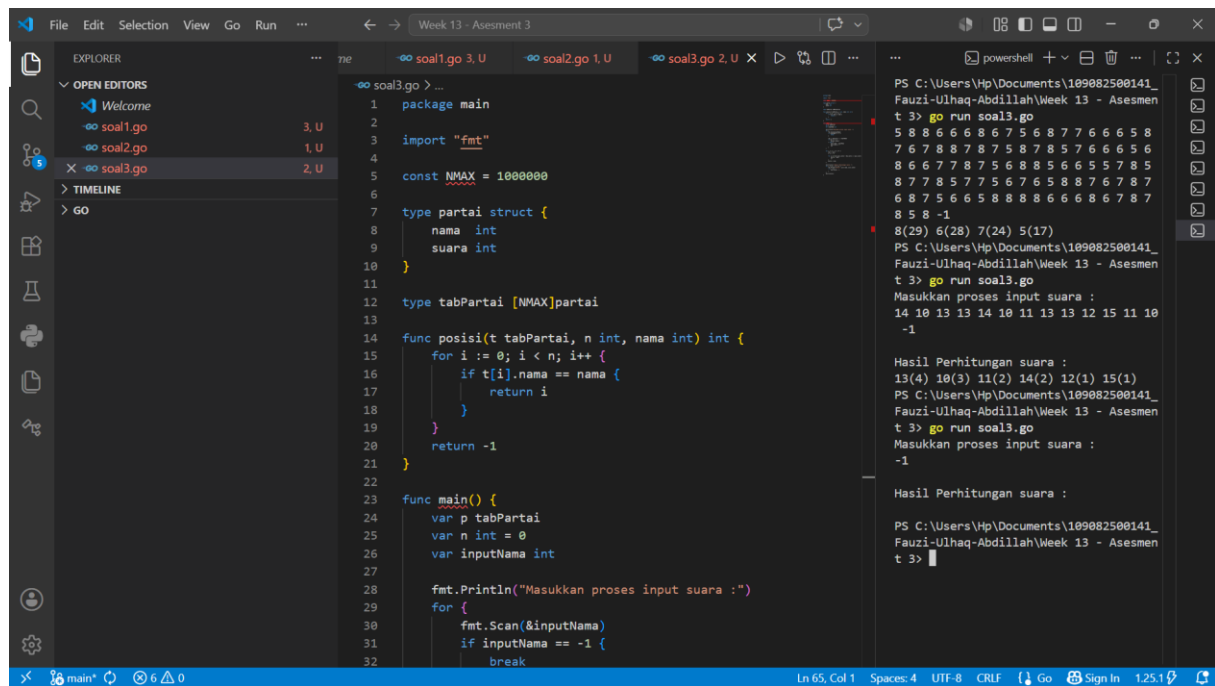
    for j >= 0 && (p[j].suara < key.suara || (p[j].suara ==
key.suara && p[j].nama > key.nama)) {
        p[j+1] = p[j]
        j--
    }
    p[j+1] = key
}

fmt.Println("\nHasil Perhitungan suara :")
for i := 0; i < n; i++ {
    fmt.Printf("%d(%d)", p[i].nama, p[i].suara)
    if i < n-1 {
        fmt.Print(" ")
    }
}

fmt.Println()
}

```

Screenshot Output :



```
1 package main
2
3 import "fmt"
4
5 const NMAX = 1000000
6
7 type partai struct {
8     nama int
9     suara int
10 }
11
12 type tabPartai [NMAX]partai
13
14 func posisi(t tabPartai, n int, nama int) int {
15     for i := 0; i < n; i++ {
16         if t[i].nama == nama {
17             return i
18         }
19     }
20     return -1
21 }
22
23 func main() {
24     var p tabPartai
25     var n int = 0
26     var inputNama int
27
28     fmt.Println("Masukkan proses input suara :")
29     for {
30         fmt.Scan(&inputNama)
31         if inputNama == -1 {
32             break
```

```
PS C:\Users\Hp\Documents\109082500141_
Fauzi-Ulhaq-Abdillah\Week 13 - Asesmen
t 3> go run soal3.go
5 8 6 6 6 8 6 7 5 6 8 7 7 6 6 5 8
7 6 7 8 8 7 8 7 5 8 7 8 5 7 6 6 5 6
8 6 6 7 7 8 7 5 6 8 8 5 6 6 5 5 7 8 5
8 7 7 8 5 7 7 5 6 7 6 5 8 8 7 6 7 8 7
6 8 7 5 6 6 5 8 8 8 6 6 6 8 6 7 8 7
8 5 8 -1
8(29) 6(28) 7(24) 5(17)
PS C:\Users\Hp\Documents\109082500141_
Fauzi-Ulhaq-Abdillah\Week 13 - Asesmen
t 3> go run soal3.go
Masukkan proses input suara :
14 10 13 13 14 10 11 13 13 12 15 11 10
-1

Hasil Perhitungan suara :
13(4) 10(3) 11(2) 14(2) 12(1) 15(1)
PS C:\Users\Hp\Documents\109082500141_
Fauzi-Ulhaq-Abdillah\Week 13 - Asesmen
t 3> go run soal3.go
Masukkan proses input suara :
-1

Hasil Perhitungan suara :
PS C:\Users\Hp\Documents\109082500141_
Fauzi-Ulhaq-Abdillah\Week 13 - Asesmen
t 3>
```

Penjelasan Singkat Cara Kerja Program :

Program tersebut bekerja dengan cara mengakumulasi setiap nomor partai yang dimasukkan oleh pengguna hingga dihentikan dengan angka -1. Setelah data terkumpul, program akan langsung menyortir dan menampilkan hasil akhirnya dari suara terbanyak, jika ada partai yang mendapat jumlah suara sama (seri), maka nomor partai yang lebih kecil akan diprioritaskan untuk tampil lebih dulu